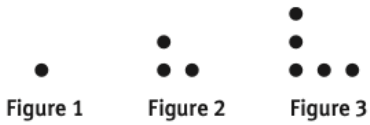
Math Course 3
VI-1 Patterns and Modeling: Representing Patterns

Name:_____
Date:_____Per:____

For problems 1 - 10, utilize the figure below:



1) Draw the 4th and 5th figures in the sequence.
Figure 4: Figure 5:

2) Organize the number of dots in each figure into the table (Column 2 ONLY)

Figure Number (*n*)	Number of Dots	Difference in # of Dots
1	**1**	—------
2	**3**	2
3		
4		
5		
6		
7		
8		
9		
10		

3) Let 'n' represent the figure number.

a) Use 'n' to write an expression that could be used to determine the number of dots in figure 'n.'

b) What value would you substitute for 'n' to determine the number of dots in the 3rd figure?

c) Use the expression to determine the number of dots in the 100th figure.

4) Subtract consecutive terms in the dots column and record this information in the final column of the table.

5) When the difference of consecutive terms is always the same it is called:

6) How does the value in the difference column relate to the pattern we observed?

- Turn Over -

Practice:

For problems 1 - 5. Use the figures below.



1) Draw the 4th and 5th figures.

2) Fill in the table for the first 6 figures.

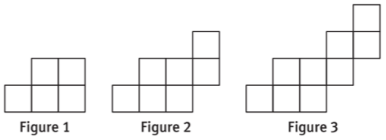
Figure Number (*n*)	**Number of Dots**	**Difference in # of Dots**
1		—-----
2		
3		
4		
5		
6		

3) What is the constant difference?

4) Let 'n' represent the figure number. Use 'n' to write an expression that could be used to determine the number of dots in figure 'n.'

5) Use your expression from problem 4 to determine how many dots would be in the 51st figure.

For problems 6 - 10, use the figures below.



6) Draw the 4th figure

7) Complete the table for the first 6 figures

Figure Number (*n*)	**Number of Squares**	**Difference in # of Squares**
1		—-----
2		
3		
4		
5		
6		

8) What is the constant difference?

9) Let 'n' represent the figure number. Use 'n' to write an expression that could be used to determine the number of dots in figure 'n.'

10) How many squares are in the 100th figure?